

BOOTSTRAP v2: orderflow-zero

A blank-slate trading research system for NSE cash equities on **Upstox**, combining **order flow, SMC, volume profile and VWAP zones**, with market context, a zone engine, pre-registered setups and honest validation.

Code is built from zero in Codex Cloud (no network, no secrets, tested against simulators). It runs on the owner's Linux machine (systemd user services), later possibly a VM in Mumbai.

Nothing in this repo places orders.

0. The idea in one paragraph

Price reacts at **zones** where several kinds of levels agree (SMC structure, volume profile, VWAP bands, prior-day levels). **Context** decides which zones matter today (index regime, sector strength, volatility, time of day). **Order flow** at the zone is the trigger that shows whether buyers or sellers are actually winning there. Each layer has one job: WHERE (zones), WHEN/WHICH (context), NOW (order flow). A trade plan needs a stop beyond the zone's invalidation and a target at the next opposing zone that is at least 3x the round-trip cost. Every idea is proven on data it was never tuned on, or it is dropped.

1. Permanent rules (copy verbatim into AGENTS.md)

Lessons already paid for

1. Every HTTP call sends an explicit User-Agent (Upstox's Cloudflare blocks Python's default urllib User-Agent with HTTP 403 / Error 1010).
2. The access token is read from a 600-permission file at connect time and re-read once on 401/403. No environment handoffs. Never log or print token values.
3. Never write to disk inside the websocket receive loop: receive -> bounded queue -> batched

- writer thread. (Inline writes caused 70 s lag and 43 disconnects in one session.)
4. Log the status code and sanitized body of every auth/connect failure.
 5. Preflight HARD gates: token missing/expired, or feed-authorize non-200 after retries. Everything else is a SOFT alert. Recording is read-only and never blocked by soft checks.
 6. One Upstox websocket client owns the feed. Others use REST or read the recorder's output.
 7. NSE holiday calendar respected by every scheduled job.
 8. Timestamps tz-aware, compared in UTC, displayed in Asia/Kolkata.
 9. Every derived value carries an availability timestamp; nothing uses data stamped after it. Structures (swings, BOS, FVG, sweeps, exhaustion) exist only after their confirming bars close.
 10. The live feed is snapshot-based: trade size, aggressor side, delta and footprint are ESTIMATES with a confidence field. Bar-based history (VWAP, volume profile) is APPROXIMATE.
 11. Gaps, resets, duplicates and out-of-order messages are flagged and counted, never dropped silently.
 12. Layers have roles, not votes. A layer stays only if a matched ablation shows it improves results on unseen data after costs.

How agents work

- Max 2 attempts or 90 minutes per problem; then `docs/stuck/<task>.md` and move on.
- Tests first: hand-built sequences with known answers for every metric and structure.
- Truncation test for everything time-based: the value at t is unchanged when data after t is removed.
- No network, keys or live calls in Cloud tasks; use the simulators.
- One task per commit ("T##: ..."); `docs/ROADMAP.md` status updated after each.
- Never weaken or delete a test to make it pass.

2. Upstox data sources (what each layer uses)

Source	Gives	Used by
Market Data Feed V3 websocket (full mode; 30-	ltp, ltq, ltt, cumulative volume (vtt), OI, best 5 bid/ask, total	order flow, tick volume profile, live VWAP, depth

level depth if plan allows)	buy/sell qty	metrics
Same feed, index instruments	Nifty 50, Bank Nifty, sector indices, India VIX ticks	context layer
Historical Candle V3	1-minute OHLCV from Jan 2022 (stocks and indices)	SMC, bar VWAP, bar volume profile, levels, context, backtests
Instrument master	instrument keys, tick size, lot size, segment	everything
NSE bhavcopy (public)	official daily OHLCV, delivery %	validation, delivery features

Facts: feed authorize returns a websocket URL; messages are protobuf (use the official SDK classes or the official .proto fetched in setup). Tokens expire daily ~03:30 IST. Concurrent websocket connections per account are limited. `vtt` is cumulative and can reset; `ltq` is only the last trade's size; snapshots can skip trades.

Order flow has no history at Upstox. It exists only from the day recording starts, so the recorder is the first thing deployed, and existing recordings are imported (see T05).

3. Architecture

```

ingest/    feed client, recorder, history downloader, importer for existing recor
decode/    protobuf -> TickEvent; candles -> Bar
trades/    per-tick volume, aggressor side + confidence
bars/      1m/5m/15m/60m time bars, volume bars, from ticks (live) or candles (hi
layers/
  orderflow/  footprint, delta, CVD, imbalances, depth, OFI, sweeps, icebergs,
              absorption, exhaustion, divergence, VPIN, Kyle's lambda
  profile/    session/developing/composite profiles, POC/VAH/VAL, HVN/LVN, naked
              initial balance
  vwap/       session VWAP + sigma bands, anchored VWAPs, weekly VWAP, VWAP even
  smc/        swings, structure, BOS/CHoCH, displacement, order blocks, FVGs,
              liquidity pools, sweeps, premium/discount, multi-timeframe bias
  levels/     prior day/week high/low/close, opening ranges, gaps
context/    index regime, India VIX regime, sector relative strength, breadth,
              time of day, liquidity tier, event-day flags
zones/      zone engine: cluster all levels into zones with components and freshne
setups/     pre-registered setups S1-S5, trade plans, cost gate

```

```

engine/    ONE code path for live and replay; per-minute point-in-time output
research/  event study, backtester (history tier), shadow signal ledger (live tie
ops/       systemd units/timers, preflight, alerts, status board, runbook, instal
ui/        read-only live view

```

All data contracts in `src/orderflow/schema.py` (pyarrow). Parameters in `config/default.yaml`.

4. Exact definitions

4.1 Order flow (live tier)

- **Per-tick volume:** $v_t = v_{tt_t} - v_{tt_{t-1}}$; negative or reset $\rightarrow 0$, flag `VOLUME_RESET`.
First tick $\rightarrow 0$, flag `FIRST_TICK`.
- **Aggressor (quote rule, tick fallback):** last quote strictly before the trade, $p = ltp$:
 $p \geq ask_1$ BUY/HIGH; $p \leq bid_1$ SELL/HIGH; $p > mid$ BUY/MEDIUM; $p < mid$ SELL/MEDIUM;
otherwise tick rule vs last different price $\rightarrow$ BUY/SELL/LOW; else UNKNOWN (reported separately).
- **Delta** = $buy_vol - sell_vol$ per bar; **CVD** from session open; $unknown_vol$ per bar.
- **Footprint:** per bar, per tick-aligned price: bid_vol (sell-aggressor), ask_vol (buy-aggressor).
- **Diagonal imbalance:** buy at p if $ask_vol(p) \geq R * bid_vol(p - tick)$ and $ask_vol(p) \geq min_vol$;
sell at p if $bid_vol(p) \geq R * ask_vol(p + tick)$ and $bid_vol(p) \geq min_vol$. $R=3.0$;
 min_vol = 20th percentile of level volume so far that day. **Stacked:** ≥ 3 consecutive levels.
- **Depth imbalance L:** $(\sum bid_qty - \sum ask_qty) / (\sum bid_qty + \sum ask_qty)$, levels 1..L.
- **OFI (Cont-Kukanov-Stoikov), level 1:**
$$e_n = 1[Pb_n \geq Pb_{n-1}]qb_n - 1[Pb_n \leq Pb_{n-1}]qb_{n-1} - 1[Pa_n \leq Pa_{n-1}]qa_n + 1[Pa_n \geq Pa_{n-1}]qa_{n-1};$$

 $OFI(interval) = \sum e_n$. **Multi-level OFI:** same per level $m=1..L$, each normalized by trailing average depth at that level; report per level and the sum.
- **Sweep (book):** best ask (bid) moves through $\geq k$ previously displayed levels in one snapshot interval with volume > 0 . $k=3$.

- **Iceberg (heuristic):** level stays at best bid/ask; traded volume there $> F \times \text{max displayed qty}$ ($F=3$) with ≥ 2 refills.
- **Absorption:** in $W=3$ bars, volume within 1 tick of a level $\geq A \times \text{median level volume}$ ($A=3$), aggressor volume dominant INTO the level, price not beyond it by > 2 ticks.
- **Exhaustion:** new $N=20$ -bar extreme with $|\text{bar delta}| \geq 90\text{th pct of the day so far}$, and no extension within $M=3$ bars (confirmed after those bars).
- **Delta divergence:** higher swing high with lower CVD high (bearish); lower low with higher CVD low (bullish); swings confirmed per 4.4.
- **VPIN:** volume buckets $V = \text{trailing 20-day avg daily volume} / 50$; per bucket $|V_{\text{buy}} - V_{\text{sell}}|/V$; mean of last 50 buckets; UNKNOWN split evenly, share reported.
- **Kyle's lambda:** rolling 30 x 1-minute OLS slope of mid return on delta, with R^2 and n .

4.2 Volume profile

- **Session profile:** volume per tick-aligned price (ticks live; history: each 1-minute bar's volume spread evenly across its high-low range, labelled APPROXIMATE).
- **POC** = max-volume price (tie $\rightarrow$ nearest session VWAP). **Value area:** expand from POC toward the side with more volume until 70% covered $\rightarrow$ VAL/VAH.
- **Developing** POC/VAH/VAL each minute from data up to that minute.
- **HVN/LVN:** local max/min of the 3-tick smoothed profile.
- **Composite profiles:** last 5 and 20 sessions.
- **Naked POC:** a prior session POC not traded since; carried forward until touched.
- **Initial balance:** high/low of the first 60 minutes; IB extensions as multiples of IB range.

4.3 VWAP zones

- **Session VWAP** = $\text{sum}(p \cdot v) / \text{sum}(v)$ (ticks live; history uses typical price $(H+L+C)/3$ x bar volume, APPROXIMATE).
- **Bands:** $\sigma = \sqrt{\text{sum } v \cdot (p - \text{VWAP})^2 / \text{sum } v}$; bands at $\pm 1, 2, 3$ sigma.
- **Anchored VWAPs** from: session open, gap open, prior-day high and low bars, each confirmed swing high/low (4.4), event times (results/announcements), week and month open.
- **VWAP events:** reclaim (close back above after ≥ 2 closes below), rejection (touch and close away), band tag (± 2 or 3 sigma touch), band acceptance (≥ 3 closes beyond a band).

4.4 SMC (timeframes 1m, 5m, 15m, 60m)

- **Swing high/low:** fractal with $N=2$ bars each side; confirmed only after N bars close.
- **Structure:** trend from the sequence of confirmed swings. **BOS:** close beyond the last swing in the trend direction. **CHoCH:** first close beyond the last swing AGAINST the trend.
Close-based by default (wick option in config).
- **Displacement:** bar range $\geq 1.5 \times \text{ATR}(14)$ and body $\geq 60\%$ of range.
- **Order block (bullish):** last down-close candle before a displacement leg that produced a bullish BOS; zone = that candle's [low, high] (or [low, open] by config). Mitigated on first return; invalidated on a close below its low. Mirror for bearish.
- **FVG (bullish):** low of bar 3 $>$ high of bar 1; gap [high1, low3]; minimum size ≥ 2 ticks and

$= 0.1 \times \text{ATR}$. Partially filled when traded into; fully filled on close below high1. Mirror for bearish.
- **Liquidity pools:** equal highs/lows (≥ 2 confirmed swings within $0.1 \times \text{ATR}$), prior day/week high/low, session high/low, IB high/low.
- **Liquidity sweep:** trade beyond a pool by ≥ 1 tick, then close back inside within $M=3$ bars
-> SWEEP_REVERSAL; closes beyond for M bars -> SWEEP_CONTINUATION (acceptance).
- **Premium/discount:** within the current dealing range (last confirmed swing low to high): above 50% premium, below discount; OTE = 62-79% retracement.
- **Multi-timeframe bias:** 60m and 15m structure direction; aligned / mixed / opposed.

4.5 Levels

Prior day high/low/close, prior week high/low, opening ranges (5/15/30 min), gap size and gap-fill status, round numbers off by default.

4.6 Context

- **Index regime:** Nifty 50 and Bank Nifty 15m structure + position vs session VWAP -> TREND_UP / TREND_DOWN / RANGE.
- **Volatility regime:** India VIX percentile over 1 year -> LOW / NORMAL / HIGH.
- **Relative strength:** stock return minus its sector index return, since open and over 5 days.

- **Breadth:** % of the universe above session VWAP; advancers/decliners.
- **Time of day:** OPEN (09:15-09:45), MORNING, MIDDAY, AFTERNOON, CLOSE (14:45-15:15).
- **Liquidity tier:** median traded value and median spread over 20 days -> tiers 1-3.
- **Event flags:** results day, announcement in the last 60 min, expiry day, budget day.

4.7 Zone engine

At each minute, collect every active level/zone (OB, FVG, POC/VAH/VAL, naked POC, HVN, VWAP

and bands, anchored VWAPs, prior-day levels, liquidity pools) as price intervals. Cluster those

within $0.25 \times \text{ATR}(14, 5m)$ into **zones**. Each zone records: price range, components by type,

count of distinct types, freshness (minutes since created), touches, the side it serves (support/resistance), and an invalidation price. Zones are LOCATION context, not votes.

4.8 Setups (pre-registered; parameters in config)

- **S1 Sweep and reclaim:** liquidity sweep reversal into a zone with ≥ 2 component types; trigger = absorption or delta flip at the zone (live tier) or reclaim close (history tier).
- **S2 Retest in discount:** OB/FVG retest in discount (premium for shorts) with aligned 15m/60m bias and VWAP on the trade side; trigger = positive OFI/delta (live) or rejection close (history).
- **S3 VWAP band reversion:** +/-2 sigma tag in a RANGE regime with exhaustion or absorption; target VWAP.
- **S4 Acceptance breakout:** acceptance beyond VAH/PDH (VAL/PDL) in a TREND regime with stacked imbalances and rising CVD (live) or 3 closes beyond (history); target next zone or IB extension.
- **S5 Naked POC magnet:** in a direction-confirmed move, target the nearest naked POC.
- **Trade plan for every setup:** stop beyond the zone invalidation + buffer ($0.1 \times \text{ATR}$); target = next opposing zone or liquidity pool; reject if reward/risk < 1.5 or target distance $< 3 \times \text{round-trip cost} + \text{spread}$. Size by volatility (risk per trade in config). Forced exit 15:15.

4.9 Validation

- **History tier (2022 onward, candles):** S1-S5 without order flow triggers, purged walk-forward, costs at 10/20/30 bps, gross and net PF, folds profitable, loss autopsy.
- **Live tier (recordings):** every setup forming live is logged in the shadow signal ledger with
zone components, context, order flow state at bar close, and outcomes at +5/15/60 min and at
the close. Formal evaluation at 20 and 60 recorded days: does the order flow trigger improve
outcomes vs untriggered setups, after costs?
- Every variant counted; holdout used once; layers added only through ablation; retired ideas
get an autopsy. Edge board: `docs/EDGE_BOARD.md`.

5. Roadmap (one commit per task; parallel groups noted)

Task	Depends	Builds	Done when
T00	-	Scaffold: AGENTS.md (section 1 verbatim), docs from sections 0-4, pyproject [dev], src/orderflow/, schema.py, config/default.yaml with every parameter	pytest green
T01	T00	Simulators: V3 protobuf feed (trend, range, gaps, resets, out-of-order, disconnect, 403, expired token, index instruments) and candle history; decoders	Round-trip and edge-case tests
T02	T01	auth (OAuth, daily refresh to token file), preflight (hard/soft, retries 08:30-09:10)	Expired/missing/401/403 tests; User-Agent always set
T03	T01	Feed client + recorder (queue, batched writer, crash-safe, lag/drop metrics, quality report), stocks + indices + VIX	Simulated 6-hour session, zero drops, receive loop never blocked
T04	T01	trades + bars (volume, aggressor, confidence; 1/5/15/60m and volume bars)	Known-answer tests

T05	T01	History downloader (Historical Candle V3 1-min from 2022, resumable, throttled, User-Agent, gap manifest, bhavcopy cross-check) + importer for existing raw recordings into this repo's schema	Simulator-backed tests; importer round-trip
T06	T04	Order flow core: footprint, delta, CVD, diagonal/stacked imbalances	Known-answer + truncation tests
T07	T04	Depth: depth imbalance, OFI, multi-level OFI, book sweeps, iceberg	Known-answer tests
T08	T06	Flow events: absorption, exhaustion, divergence, VPIN, Kyle's lambda	Known-answer tests; confirmation timing
T09	T04	Volume profile: session, developing, composite 5/20, naked POC, HVN/LVN, IB	Known-answer + truncation tests
T10	T04	VWAP: session + bands, anchored VWAPs, weekly/monthly, VWAP events	Known-answer + truncation tests
T11	T04	SMC: swings, structure, BOS/CHoCH, displacement, OB, FVG, pools, sweeps, premium/discount, MTF bias	Known-answer tests per structure; confirmation timing
T12	T04	Levels: prior day/week, opening ranges, gaps	Known-answer tests
T13	T03, T04	Context: index regime, VIX regime, sector RS, breadth, time of day, liquidity tier, event flags	Known-answer tests
T14	T09-T12	Zone engine	Clustering, freshness and invalidation tests
T15	T08, T13, T14	Setups S1-S5, trade plans, cost gate, sizing	Each setup triggers on a crafted scenario and not on its near-miss
T16	T15	Engine: one live/replay path, per-minute point-in-time output, events and setup tables	Deterministic replay; live simulation equals replay
T17	T16	Research: history-tier backtester (purged walk-forward, costs), event study, shadow	End-to-end on simulated data with

		ledger, edge board, variant ledger	report templates
T18	T02, T03, T05	Ops: systemd units/timers (refresh 03:45, preflight 08:30, recorder 09:00-15:35, history nightly), holiday calendar, Discord alerts, status board, RUNBOOK.md, install script	Install dry-run on a clean container
T19	T16	Read-only live view: per symbol, zones, VWAP bands, profile, footprint/CVD, recent events and setups	Renders from replay output

Parallel groups: [T02, T03, T04, T05] after T01; [T06, T07, T09, T10, T11, T12] after T04; [T08, T13] next; then T14 -> T15 -> T16 -> [T17, T18, T19].

Deploy early: after T02 + T03 + T18, the owner installs the recorder so live data starts accumulating while the layers are built.

6. Codex Cloud environment

Setup script (`scripts/codex_setup.sh`):

```
set -euo pipefail
python -m pip install --upgrade pip
pip install -e ".[dev]" || pip install pytest hypothesis pyarrow polars numpy pro
pip install upstox-python-sdk || true
# If the SDK lacks the V3 feed protobuf, download the official MarketDataFeedV3.p
# Upstox's documentation into proto/ and compile it with grpcio-tools here.
```

Agent internet OFF (or Upstox documentation domains only, GET). No secrets.

7. Owner steps on the machine (after T18)

1. Clone, run the install script.
2. Put API key/secret/redirect URL and the Discord webhook in the files RUNBOOK.md names (600 perms).
3. Do the first OAuth login once.
4. Stop the old recorder first (one feed client per account), then start the new recorder.
5. Import existing recordings (T05 importer) so no recorded days are lost.

6. Run replay on recorded days; enable live research outputs after replay checks pass.